

Chloride

Test Details

METHODOLOGY

Ion-selective electrode (ISE)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate electrolytes, acid-base balance, water balance. Chloride generally increases and decreases with plasma or serum sodium.

Chloride is **increased** in dehydration, with ammonium chloride administration, with renal tubular acidosis (hyperchloremic metabolic acidosis) and with excessive infusion of normal saline. Differential diagnosis of acidemias and alkalemias. Chloride is higher in hyperparathyroidism than in some of the other causes of hypercalcemia, but a great deal of overlap exists.

Chloride is **decreased** with overhydration, congestive failure, syndrome of inappropriate secretion of ADH, vomiting, gastric suction, chronic respiratory acidosis, Addison disease, salt-losing nephritis, burns, metabolic alkalosis, and in some instances of diuretic therapy.

Custom Additional Information

Like other electrolytes, chloride cannot be interpreted without clinical knowledge of the patient. A diagnostic approach to the evaluation of hyperchloremic metabolic acidosis includes use of the urinary anion gap in conjunction with measurement of plasma potassium and urinary pH.¹

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

96–106 mmol/L

*Reference interval has been verified for all age groups including pediatrics (birth to 18 years) and geriatrics.

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Improper labeling

Footnotes

1. Batlle DC, Hizon M, Cohen E, Gutterman C, Gupta R. The use of the urinary anion gap in the diagnosis of hyperchloremic metabolic acidosis. *N Engl J Med*. 1988 Mar 10; 318(10):594-599. PubMed 3344005
(<http://www.ncbi.nlm.nih.gov/pubmed/3344005>)